

Niharika Sathyanarayana

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EDUCATION

Master of Science in Computer Science

Dec 2024

University of Colorado Boulder | Boulder, CO

CGPA: 3.84/4.0

Coursework: Datacenter Scale Computing System, Natural Language Processing, Big data Architecture and Analytics, Cybersecurity

Bachelor of Engineering, Computer Science Engineering

Jul 2022

Bangalore Institute of Technology affiliated to Visvesvaraya Technological University | Bangalore, India

CGPA: 8.94/10.0

Coursework: Data Structures and Algorithms, Object Oriented Programming, Database, Computer Networks, UNIX

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, Bash, C, C++, SQL, HTML/CSS, PyTorch
- **Frameworks & Libraries:** Django, Flask, FastAPI, AngularJS, React, Bootstrap, Spring Boot, RESTful API, Node.js
- **Tools:** Google Cloud, AWS, Git, Jira, Figma, Jenkins, CI/CD, Docker, Terraform, Kubernetes, OpenAPI
- **Databases & Big Data:** MySQL, PostgreSQL, MongoDB, Firebase, Spark, Kafka
- **Certifications:** Microsoft Technology Associate in Python Programming

EXPERIENCE

Virufy | Software Engineer | Remote

Feb 2025 – Present

- Built scalable RESTful APIs using **Python, Django, and FastAPI**, integrated with **AWS**-hosted services to support secure respiratory diagnostic workflows
- Implemented **JWT-based auth** and HTTPS protocols, ensuring compliance with secure data access and SOA principles
- Automated **CI/CD pipelines** using **GitHub Actions and Docker**, followed TDD practices with **PyTest** and **Postman** for validation

University of Colorado Boulder (HIRO LAB) | Research Assistant | Boulder, CO

Jan 2024 – Dec 2024

- Developed Bilateral Segmentation and Disparity Refinement (**BSDR**), a lightweight neural network for real-time dense spatial perception in low-SWaP robots, achieving **11 FPS** and improving occlusions and object recognition through enhanced 3D data curation and annotation using CREStereo

Rakuten | Software Engineer Intern | Bangalore, India

Jan 2022 – Jul 2022

- Developed and enhanced key B2B order management portal features using **AngularJS, HTML, CSS, TypeScript Spring Boot, Postman and MySQL**. Collaborated with cross-functional teams via JIRA and implemented **CI/CD** pipelines, resulting in a **20%** increase in feature deployment speed and a **25%** reduction in development cycles
- Improved application performance by resolving **35** high-priority bugs, reducing post-release defects by **20%** through comprehensive testing with **JUnit**, and redesigning the website to enhance user experience, increasing engagement by **30%**

Defence Research & Development Organisation(CAIR) | Software Development Intern | Bangalore, India

Aug 2021 – Dec 2021

- Built a secure data-sharing platform with **RESTful APIs** and **PostgreSQL**, implementing **SSL/TLS** encryption, **OAuth2** role-based access control, and **JWT** authentication, significantly enhancing data protection
- Developed responsive interfaces using **React** and **Bootstrap**, improving research collaboration and reducing time to insights by **30%**. Streamlined deployments with **Jenkins** CI/CD pipelines, following Test-Driven Development practices
- Led code management using **Git** within an **Agile** framework, enhancing collaboration and reducing conflicts by **15%**

PROJECTS

StockWatch

Jan 2023 – May 2023

- Designed and executed a high-performance Big Data pipeline using **Kafka, Pyspark, MongoDB, and AWS**
- Successfully processed a substantial data load, handling **1000** tweets per hour from NYSE and NASDAQ stocks via Twitter and Yahoo APIs, all with a remarkable processing latency of **< 500** milliseconds
- Implemented sentiment analysis with **NLTK** and **spaCy** to correlate tweet sentiment with stock prices

Movie Recommendation System

Aug 2022 – Dec 2022

- Developed a Collaborative filtering-based movie recommendation system using **FastAPI, MongoDB, and ReactJS**
- Implemented **Docker** containerization for expedited deployment, resulting in a reduction in deployment times by **25%**
- Leveraged **Google Cloud Platform (GCP)** for hosting, ensuring efficient scalability to enable a **35%** concurrent user surge

Automatic Hate Speech using Ensemble Method and NLP Techniques | Final year project at BIT

Oct 2021 – Jul 2022

- Led a team in developing an efficient social media hate speech detection system, classifying code-mix (Hindi and English) hate speech into multi-class categories, i.e. hate, non-hate and offensive texts using soft voting
- Orchestrated ensemble integration (**Random Forest** and **Support Vector Classifiers**) with **Python, Scikit-Learn, and TF-IDF vectorization**, achieving a remarkable **90.7%** accuracy – a **9.3%** improvement over traditional methods
- Published findings in the [IEEE conference on NMITCON-2023](#) and was honoured with the **best project** award at BIT

PUBLICATIONS/ACTIVITIES

- “Deep Learning-Based Skin Lesion Detection and Classification: A Review” in [IRJET Volume 9, Issue 7](#)
- “A Review on Grape Disease Detection and Classification using Image Processing” in [IRJET Volume 8, Issue 9](#)
- Teaching Assistant at CU Boulder. Led recitations, office hours and coursework design in **C++** for 48 students for 5 semesters.